



An Interpretable Machine Learning Method in Symbolic Regression - Jessamine

Background

Black box ML models are being used for high stakes decision-making throughout society, causing problems in healthcare, criminal justice, and in other domains.

Interpretable ML is crucial for

- Ensuring transparency and accountability
- Identifying and mitigating bias
- Detecting flaws in models and data
- Combining model predictions with external information

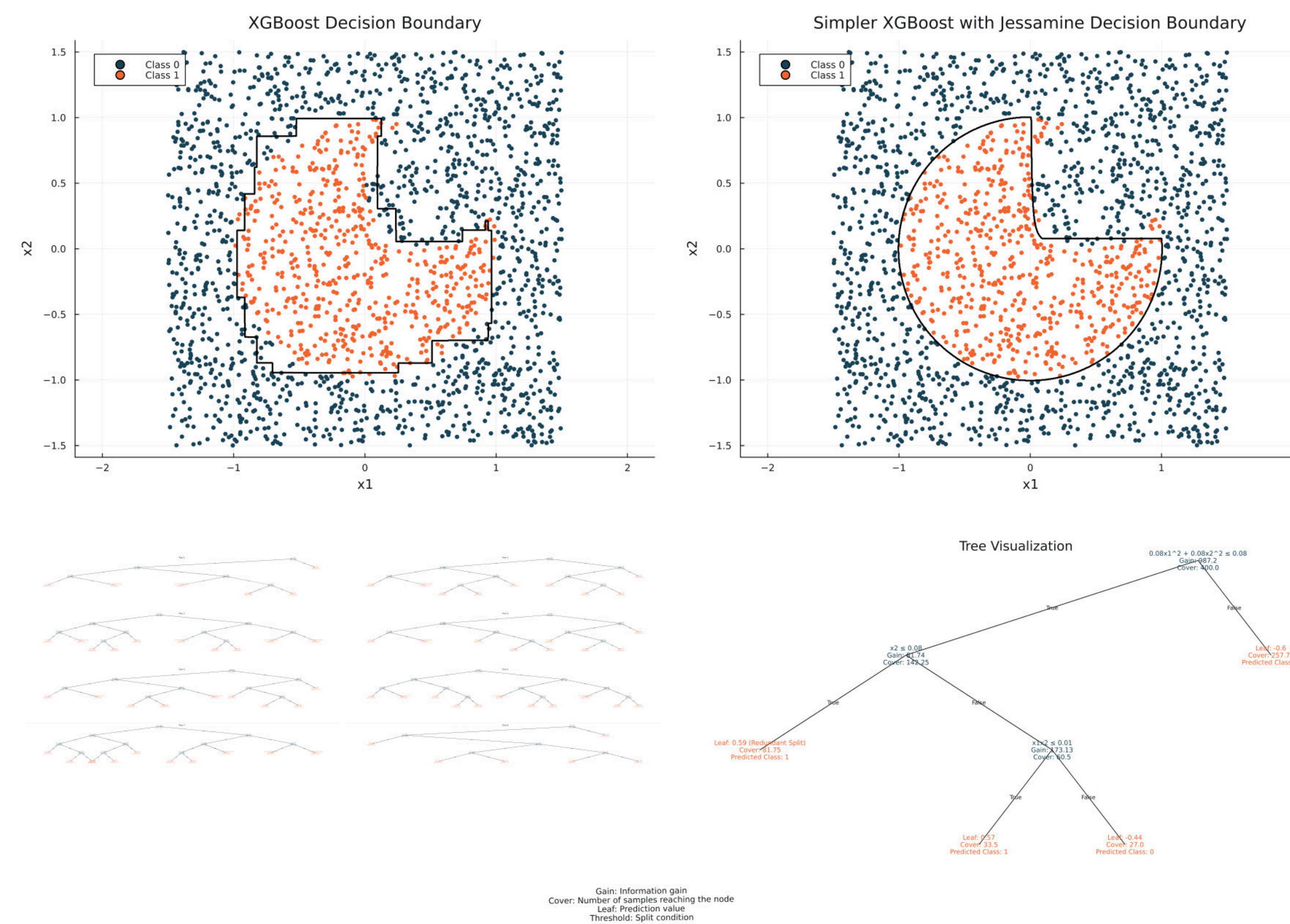
Interpretable models can match the accuracy of black box models while providing clear explanations of their reasoning process.

Jessamine

Symbolic Regression searches for a function $f(x)$ with a simple expression to model input data x and response y . Deep Neural Networks model $f(x)$ via uninterpretable nested activation functions. Decision trees are unable to capture algebraic relationships between features.

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XGBoost without vs with Jessamine



Jessamine finds transformed input features to feed into XGBoost in order to capture non-linear relationships. The plot demonstrates XGBoost limitation - requiring an uninterpretable design using 8 decision trees, 4 depth, and a complex decision boundary that will not generalize as well.

Objective Function

$$\|y - \hat{y}\|^2 + \lambda_p \|p\|^2 + \lambda_0 R$$

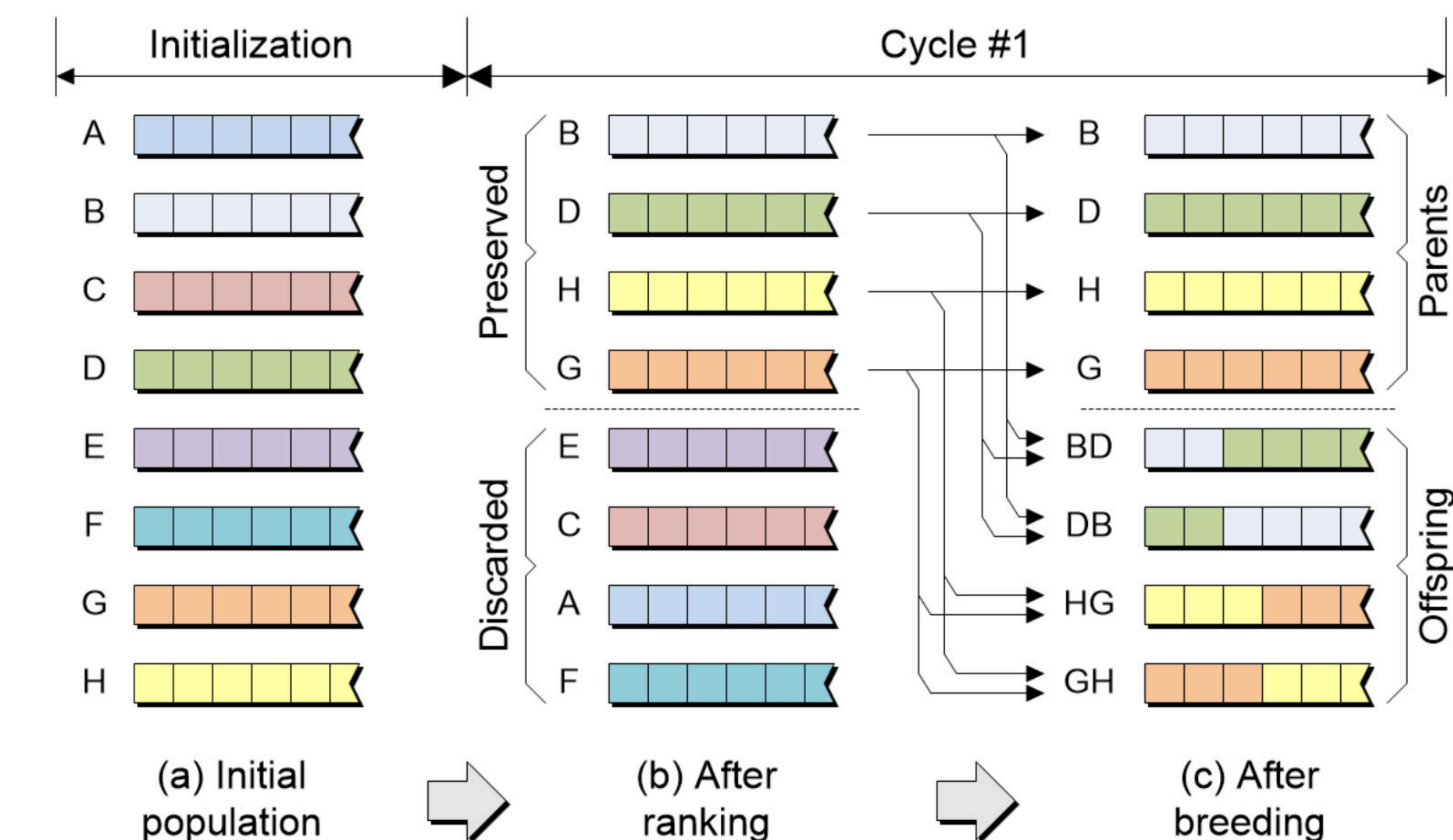
The Jessamine package optimizes a parameter vector p by minimizing the given objective function.

- $\lambda_p, \lambda_0 > 0$: Regularization parameters for controlling the magnitude of the parameter vector p and the complexity of the model, respectively.
- R : Represents the total number of operations across all instructions in the genome.

Jessamine Output

$$\begin{bmatrix} x_2 \\ x_1 x_2 \\ 0.08x_1^2 + 0.08x_2^2 \\ 0.08x_1^2 \end{bmatrix}$$

Genetic Algorithm Framework



Each genome in the population represents a function. After we iterate through populations mutating and combining the best ranked genomes, we select best and run a simplification process.

Heart Disease Real-World Dataset Prediction Accuracies - 303 People

Model	Sensitivity	Specificity	F1 Score	AUC Score
Decision Tree	0.72	0.97	0.82	0.73
Random Forest	0.72	0.97	0.82	0.93
Logistic Regression	0.79	0.94	0.85	0.95
XGBoost	0.72	0.97	0.82	0.91
XGBoost (Tuned)	0.72	0.97	0.82	0.94
Neural Network	0.69	0.97	0.80	0.95
Jessamine	0.786	0.848	0.800	0.9037

Jessamine underperforms in terms of accuracy for some real-world datasets as it is still in the early stages of development. We found the most important predictors are Oldpeak (ST depression induced by exercise relative to rest, indicating heart disease severity), MaxHR (maximum heart rate achieved during exercise, reflecting cardiovascular fitness), and ExerciseAngina (exercise-induced chest pain, a strong indicator of coronary artery disease).